

6(a)(i)	up up down	B1
6(a)(ii)	up down down	B1
6(a)(iii)	(alpha-particle is) 2 protons and 2 neutrons	C1
	6 up, 6 down	A1
6(b)(i)	most of an <u>atom</u> is empty space or the nucleus (volume) is (very) small <u>compared with the atom</u>	B1
6(b)(ii)	the nucleus is charged	B1
	the majority of the mass of atom is in the nucleus	B1
6(c)	$F = Eq$ and $a = F / m$	C1
	$a = Eq / m$ ratio = $(e / m) / (2e / 4m)$ $= 2$	A1